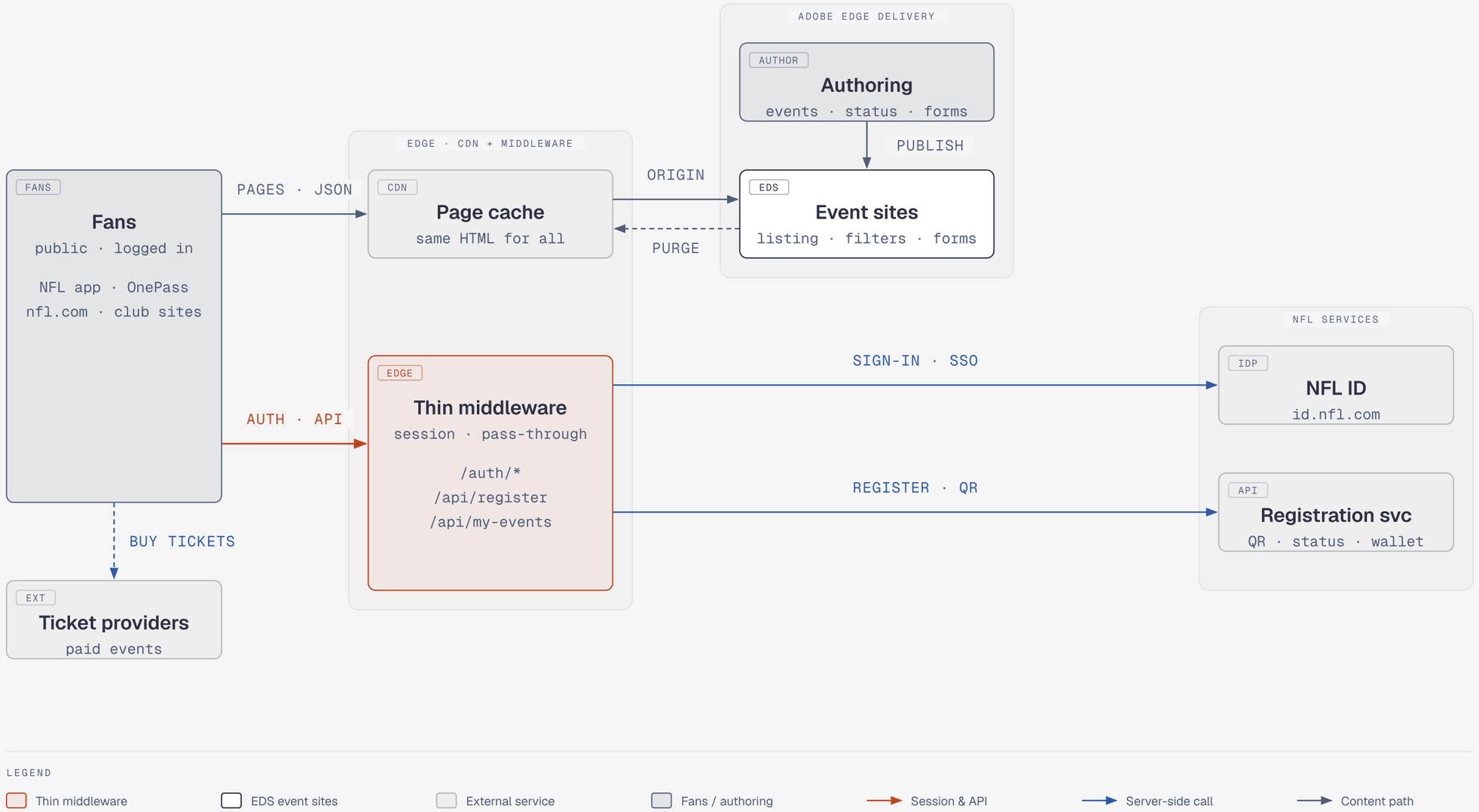
SOLUTION ARCHITECTURE · NFL EVENTS HUB ON AEM EDGE DELIVERY SERVICES

NFL Events Hub — solution architecture



01 Summary

The Events Hub is a set of fast, cached event sites on **AEM Edge Delivery Services (EDS)**, authored by clubs and the league, shown on nfl.com, club sites, the NFL app and OnePass. A **thin edge middleware** owns the signed-in session with NFL ID and forwards the few per-fan calls (register, my events). NFL's own systems stay the source of truth for identity, registrations, notifications and ticketing. The hub never takes payments: paid events link out.

Static first. Anything the same for every fan is an EDS page or JSON from the CDN.

Thin middleware. Edge worker, no database, no page rendering, no business rules beyond auth and forwarding.

No money in the hub. No prices as data, no card forms. Buy = exit link.

Direct by default. The browser talks to EDS, the CDN, push and ticket links directly. The middleware is used only for session and secrets.

NFL owns the data. Registrations, capacity, QR validity and fan contact live in NFL services, not in EDS or the worker.

One build, many surfaces. Same pages for web, embeds and in-app (app mode).

02 Business requirements → solution

REQUIREMENT	HOW IT IS MET	COMPONENT
Simple design, NFL branding	NFL Global Design Playbook as EDS design tokens; small block library	EDS
Club-managed international 1:1 events first	Per-club, per-market authoring workspaces; one codebase serves the hub and every club or event site (repoless)	EDS + authoring
Event details, venue location	Event page: details, schedule, venue and map, FAQ, sponsors	EDS blocks
All events, filters, search	Query index publishes every event's metadata as `events.json`; listing block filters by club, market, city, date, type, free/paid in the browser	EDS
Public view and logged-in view	Same cached page; blocks call `/api/me` and switch to personalized state	EDS + middleware
OnePass-style Login and Register	Both open hosted id.nfl.com (sign-in / sign-up) — identical look	NFL ID + middleware
Register for events with extra fields	Form block; fields authored per event in a sheet; submit with the fan's NFL ID	EDS + middleware + registration svc
Free and paid events	Per-event *access type* decides the button: Register, Register + time slot, Enter, Buy Tickets, Premium	EDS metadata
Buy Tickets outside the hub	Authored ticket URL and provider per event (Ticketmaster, NFL ticket account, club, On Location)	Exit link
No pricing, no credit cards	Not modelled; no payment code anywhere	—
My registered events, status	`/api/my-events` via middleware to the registration service	Middleware
QR codes and gate entry	Registration svc issues a signed QR (per-club setting) and validates it at entry	Registration svc
Add to Wallet	Signed Apple / Google Wallet pass returned with the registration	Registration svc
Real-time cancellations	Authors set status in the status sheet and publish; CDN purged in seconds; registration svc emails / pushes registrants	EDS + registration svc
Embed in NFL app, OnePass, websites	App mode (WebView, JS bridge) and embed mode (nfl.com, club sites)	EDS
Future: league events, Leap features	Same blocks for league events; Leap (badges, scanning, offers, sponsors) behind the middleware	Future

03 Components

COMPONENT	RESPONSIBILITY	TECHNOLOGY
Event sites	Hub, club and market sites, event pages, listings, forms, My Events shell	AEM EDS, repoless multi-site, vanilla JS blocks
Authoring	Clubs and league author events, form fields, status sheet; publish = live	Document authoring (Drive/SharePoint/DA) or Universal Editor; folder permissions per club
Page cache	Serves identical HTML/JSON to everyone; push-invalidated on publish	BYO CDN (Cloudflare/Fastly/Akamai) in front of EDS
Thin middleware	NFL ID session (HttpOnly cookie); pass-through to the registration service; rate limiting	Edge worker on the same CDN
NFL ID	Login, sign-up, SSO across nfl.com, NFL app, OnePass	id.nfl.com (Gigya / SAP CDC per NFL docs)
Registration service	Registrations, capacity, one-per-fan rules, QR issue and validation, wallet passes, alerts to registrants	NFL / OnePass backend (preferred); POC fallback: small serverless store
Ticket providers	Paid sales	Ticketmaster, NFL ticket account, club ticketing, On Location
Leap (future)	Badges, scanning, offers, sponsor integrations	Later, behind the middleware

04 Key flows

1. **Browse.** Fan opens hub or event page → CDN serves cached EDS HTML; listings read `events.json` (query index); status banner reads `status.json`.
2. **Log in / Register account.** Button → `/auth/login` or `/auth/signup` → id.nfl.com → back to `/auth/callback` → middleware sets session cookie. Already signed in on nfl.com = no prompt. In apps: the app hands over its session (one-time code) — no second login.
3. **Register for an event (free).** Register → form block (fields from the event's sheet) → `POST /api/register` → registration svc checks capacity and rules → confirmation, status and QR.
4. **Buy tickets (paid).** Buy Tickets → authored provider URL in a new tab. Nothing stored in the hub.
5. **My events.** `/api/my-events` → registered events, live status, QR, Add to Wallet (signed pass from the registration service).
6. **Entry.** The fan shows the QR (or wallet pass) at the venue; it is validated against the registration service.
7. **Cancellation.** Authors mark the event (or every event in a city) cancelled in the status sheet and publish → CDN purged → every page and listing shows it within seconds; the registration service emails / pushes registered fans.
8. **In-app and embeds.** Same URLs in app mode (no site chrome, native share/back/wallet via JS bridge) or embed mode on nfl.com and club sites; login from an embed opens id.nfl.com top-level.

05 Event content model (authored)

FIELD	EXAMPLE	USED FOR
club · market · language	raiders · de · de-DE	Workspaces, listing filters, localization
title · date/time · timezone	Raiders Watch Party Berlin · 2026-11-08 19:00 · Europe/Berlin	Page, listing, calendar
venue · address · geo	Kulturbrauerei · 52.539, 13.413	Venue block and map
access type	free-register · free-session · free-open · giveaway · paid-external · hospitality	Which button(s) show
registration form	link to the event's form sheet	Form block fields
capacity · time slots · age limit	500 · 18:00/19:00 · 18+	Passed to registration svc
qr required	yes / no (per club)	QR issue and scanner
ticket url · provider	https://… · Ticketmaster	Buy Tickets exit link
status	scheduled · live · cancelled · ended (from status sheet)	Banner, listing badge

06 Who owns what

DATA	SOURCE OF TRUTH	HUB KEEPS
Event content	EDS (club/league authors)	Cached pages, JSON
Fan identity	NFL ID	Session cookie only
Registrations, capacity, QR validity, check-ins	Registration service	Nothing (pass-through)
Event status	EDS status sheet (authored)	Cached status.json
Fan contact and consent	NFL CRM / notifications	Nothing
Tickets and payments	Ticket providers	Nothing

07 Non-functional

Scale. Spikes (registration opening, Draft and Super Bowl week) hit the CDN cache; only per-fan calls reach the middleware. Rate-limit and queue `/api/register`; load-test it before peak events.

Performance. EDS pages target Lighthouse 100; personalization loads after first paint.

Security. Tokens never in the browser; HttpOnly, Secure, SameSite cookies; secrets only in the worker.

Privacy. International events mean GDPR and local rules: consent for push and email, minimal form fields, data stays in NFL systems.

> **Localization.** Site per market/language from one codebase.

> **Embeds.** Allow-list embedding domains; logged-in features open top-level because browsers block third-party cookies.

> **Accessibility.** WCAG 2.2 AA blocks; QR also available as a code fans can read out.

08 Phasing

PHASE	SCOPE
POC	Hub + 3–5 club events; public and logged-in views; NFL ID login; Register with per-event form; My Events with QR; Buy Tickets exit for 1–2 events; app mode in OnePass
Phase 1 — club international events	All clubs and markets, localization, capacity and time slots, QR entry, Add to Wallet, cancellation alerts, embeds on club sites
Phase 2 — league events	Draft, Super Bowl week, international games fan events; giveaways; hospitality links
Phase 3 — Leap features	Badges, scanning rewards, offer redemption, sponsor integrations

09 Open questions

1. Does an NFL / OnePass registration API exist (capacity, QR, status), or does the POC need its own store?
2. Can the hub be registered with NFL ID (Gigya site group or OIDC client), and can OnePass and the NFL app hand over sessions?
3. Must QR codes work with existing OnePass / Leap scanners at venues?
4. Which notification channel (email, app push) should alert registrants about cancellations?
5. Wallet: does NFL have Apple pass-type and Google issuer accounts?
6. Exact ticket URLs and providers for the paid POC events.
7. Authoring choice (documents vs Universal Editor) and per-club permissions.
8. CDN choice for production (BYO CDN with edge worker vs AEM Edge Functions).